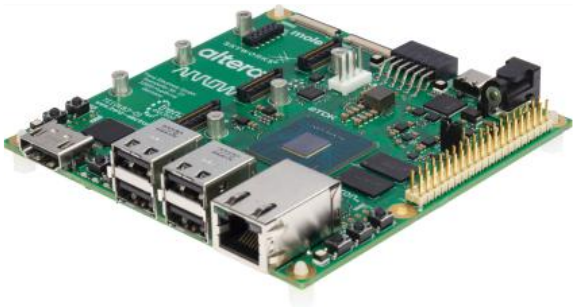


Arrow FPGA SoC Development kit
Altera™ Agilex™ 5 E-series

The Arrow AXE5-Falcon Development Kit is based on the Altera™ Agilex™-5 SoC FPGA which is manufactured in Intel 7 technology process. The Agilex-5 is a mid-range FPGA with logic sizes ranging from 50k to 656k LE which are scalable across multiple packages. The Agilex-5 features the industry's first Enhanced DSP element with AI Tensor Block, 17G transceivers, and hardened PCIe Gen4 and 10GbE IP. The Processor system offers asymmetric application processors consisting of dual Arm® Cortex® A55 and dual Cortex A76 cores (Only dual A55 on this kit). It offers Hard Memory Controllers for DDR4/LPDDR4/LPDDR5 memory, and 1.8V/2.5V/3.3V HVIO and 1.1V/1.2V/1.3V HSIO IO interfaces.
Arrow's AXE5-Falcon Dev kit enables the user to evaluate the features of the Altera Agilex-5 E-series and speed up development. With the various I/O interface connectors, the user can add peripheral expansions. The AXE5-Falcon has various push buttons, DIP Switches, RGB LEDs, and expansion connectors.
Arrow provides a User Guide, demo designs, reference designs including Linux support, on a GitHub Wiki (Scan QR code).
The FPGA is configured via the built-in USB Blaster III.



Key Features: Arrow AXE5-Falcon Kit

- A5EE013BB23BE6SCS
  - * 138 kLE
  - * Dual A55 SoC
  - * 376 multipliers 18x19
  - * 4.93 Peak INT8 (TOPS)
  - * 6.99 Mb (M20K) internal memory
  - * MIPI D-PHY interface
  - * LVDS pairs at 1.250 Mbps
  - * Secure Device Manager
- BBGA 820-pin 23mm x 23mm
- DDR4: 2x 1Gbit x16 (1Gb x 32)
- USB 2.0
- 1x 1Gb Ethernet
- HDMI Output Interface
- MicroSD Card
- QSPI Config Flash: 256 Mbit
- EEPROM
- Raspberry Pi Hat Header
- 4x RGB LEDs, Push Buttons, DIP switches
- 1x CRUVI LS + 1x PMOD connectors
- 3x CRUVI HS connector
- 2x MIPI CSI 22-pin connectors
- On-board USB Blaster III Programmer
- 5V Power via USB-C or Barrel Jack
- Size: 10cm x 10cm

Ordering number: AXE5-Falcon
Technical details: GitHub page

Partners:

altera An Intel Company
micron
samtec
SKYWORKS
TDK
trenz electronic